

MACROEVOLUTION

Module map

- Module 14: Macroevolution
- Connected lab: lab-14_macroevolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 14

Macroevolution

1

Species
concepts

2

Reproductive
isolation

3

Prezygotic
barriers

4

Postzygotic
barriers

5

Speciation
patterns
and
phylogeny

Lab evidence

lab-14_macroevolution.md

MACROEVOLUTION

Learning objectives

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Classify examples as prezygotic or postzygotic barriers.
- Compare allopatric and sympatric speciation.
- Interpret speciation as population-level divergence over time.

OBJECTIVE 1

State the biological species concept and its limits.

OBJECTIVE 2

Explain reproductive isolation as reduced gene flow.

OBJECTIVE 3

Classify examples as prezygotic or postzygotic barriers.

OBJECTIVE 4

Compare allopatric and sympatric speciation.

OBJECTIVE 5

Interpret speciation as population-level divergence over time.

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Topic sequence

- Species concepts: Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
- Reproductive isolation: Reproductive isolation reduces gene flow between populations.
- Prezygotic barriers: Prezygotic barriers prevent mating or fertilization before a zygote forms.
- Postzygotic barriers: Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.
- Speciation patterns and phylogeny: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

01

Species concepts

Species concepts define lineages using reproductive, morphological, or evolutionary criteria.

02

Reproductive isolation

Reproductive isolation reduces gene flow between populations.

03

Prezygotic barriers

Prezygotic barriers prevent mating or fertilization before a zygote forms.

04

Postzygotic barriers

Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.

05

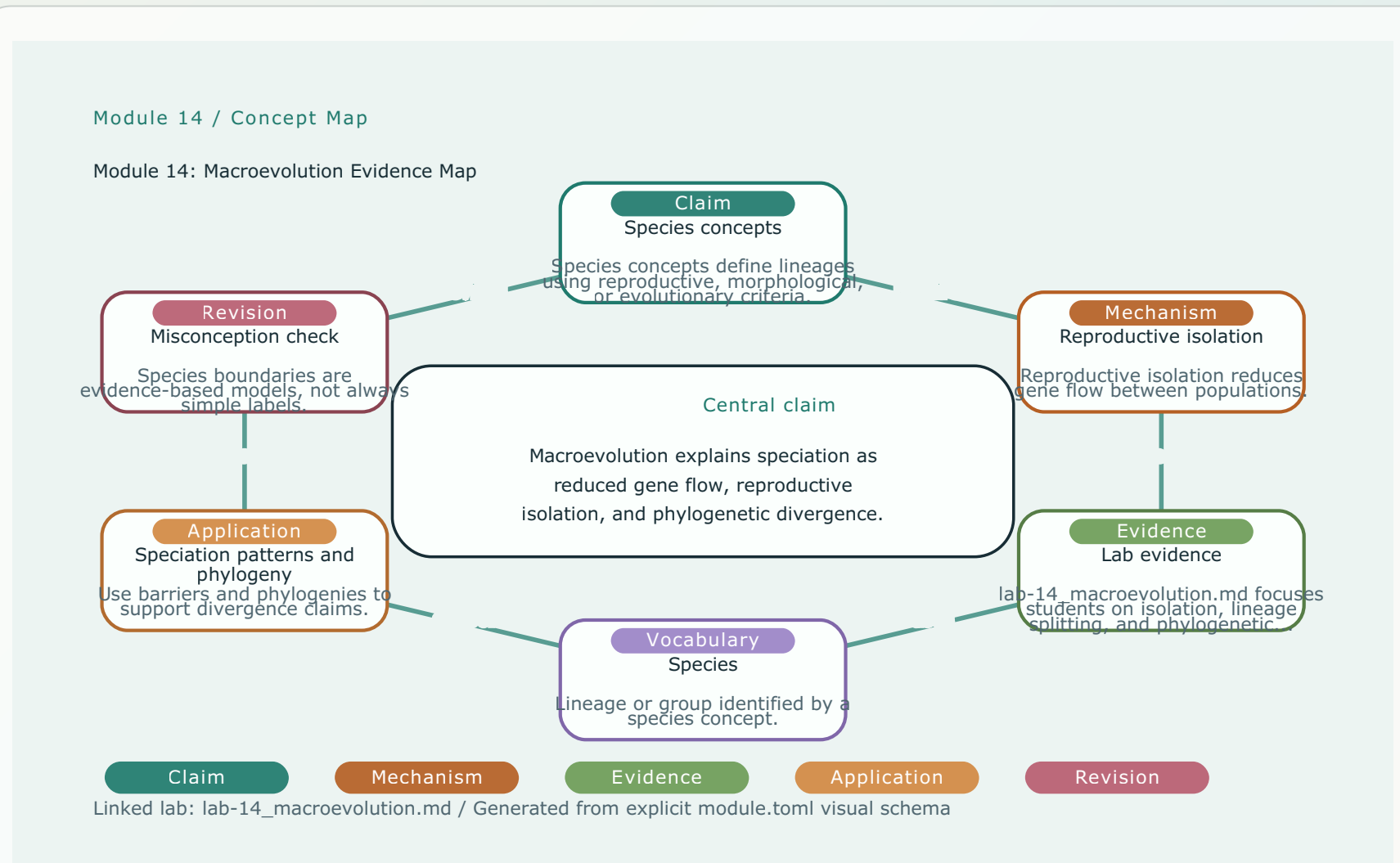
Speciation patterns and phylogeny

Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

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Module 14: Macroevolution Evidence Map

- Module focus: Macroevolution
- Central claim: Macroevolution explains speciation as reduced gene flow, reproductive isolation, and phylogenetic divergence.
- Key nodes to connect: Species concepts, Reproductive isolation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection:
lab-14_macroevolution.md supplies an evidence surface for the map.



Teaching note: Ask students to explain one edge in their own words before moving on.

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Module 14: Macroevolution Reasoning Sequence

- Module focus: Macroevolution
- Trace the model: Species concepts -> Reproductive isolation -> Prezygotic barriers -> Postzygotic barriers
- Inputs become outputs: Species concepts, Reproductive isolation -> State the biological species concept and its limits., Explain reproductive isolation as reduced gene flow.
- Feedback check: lab-14_macroevolution.md evidence checks whether students can explain prezygotic barriers.
- Lab connection: lab-14_macroevolution.md gives students a process to observe or test.

Module 14 / Process Model

Module 14: Macroevolution Reasoning Sequence

Inputs

- Species concepts
- Reproductive isolation
- Evidence from lab-14_macroevolution.md

Outputs

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Interpret speciation as population level divergence over...

Reasoning sequence

1

Species concepts

Species concepts define using reproductive, morphological, or evolutionary criteria.

2

Reproductive isolation

Reproductive isolation reduces gene flow between populations.

3

Prezygotic barriers

Prezygotic barriers prevent mating or fertilization before a zygote forms.

4

Postzygotic barriers

Postzygotic barriers reduce survival or fertility after zygote forms.

5

Speciation patterns and phylogeny

Speciation can occur allopatrically or sympatrically and is represented in phylogenetic trees.

Feedback

lab-14_macroevolution.md evidence checks whether students can explain prezygotic barriers.; Retrieval questions reveal whether terms are...

Constraint

Species boundaries are evidence-based models, not always simple labels.; Speciation can occur allopatrically or sympatrically and is represented...

Linked lab: lab-14_macroevolution.md

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Terms as evidence handles

- Species: Lineage or group identified by a species concept.
- Biological species concept: Species defined by interbreeding and reproductive isolation.
- Reproductive isolation: Barriers that reduce gene flow between groups.
- Prezygotic barrier: Barrier before fertilization.
- Postzygotic barrier: Barrier after fertilization.
- Allopatric speciation: Speciation with geographic separation.

SPECIES

Lineage or group identified by a species concept.

BIOLOGICAL SPECIES CONCEPT

Species defined by interbreeding and reproductive isolation.

REPRODUCTIVE ISOLATION

Barriers that reduce gene flow between groups.

PREZYGOTIC BARRIER

Barrier before fertilization.

POSTZYGOTIC BARRIER

Barrier after fertilization.

ALLOPATRIC SPECIATION

Speciation with geographic separation.

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Lab connection

- Lab file: lab-14_macroevolution.md
- Lab evidence should test or illustrate: Prezygotic barriers
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Species concepts

EVIDENCE

lab-14_macroevolution.md

CLAIM

State the biological species concept and its limits.

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Contrast check

- Do not stop at: Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
- Stronger explanation: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Species concepts define lineages using reproductive, morphological, or evolutionary criteria.

DEEPER

Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

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Module 14: Macroevolution Retrieval and Lab Check

- Module focus: Macroevolution
- Retrieval prompt: What does the biological species concept emphasize?
- Answer check: State the biological species concept and its limits.
- Required terms: Species, Biological species concept, Reproductive isolation
- Lab connection:
lab-14_macroevolution.md;
lab-14_macroevolution.md supplies evidence for isolation, lineage splitting, and phylogenetic pattern evidence.

Module 14 / Retrieval Card

Module 14: Macroevolution Retrieval and
Lab Check

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

What does the biological species concept emphasize?

Check: State the biological species concept and its limits.

Why does reproductive isolation matter for speciation?

Check: Explain reproductive isolation as reduced gene flow.

What is a prezygotic barrier?

Check: Classify examples as prezygotic or postzygotic barriers.

What is a postzygotic barrier?

Check: Compare allopatric and sympatric speciation.

Terms to use

Species, Biological species concept, Reproductive isolation, Prezygotic barrier, Postzygotic barrier

Lab connection

lab-14_macroevolution.md supplies evidence for isolation, lineage splitting, and phylogenetic pattern evidence.

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Under the biological species concept, two groups count as different species when they:
- 2. Two related species breed at different times of year and so never mate. This is an example of a:
- 3. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
- 4. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

QUESTION 1**Key: C**

Under the biological species concept, two groups count as different species when they:

QUESTION 2**Key: A**

Two related species breed at different times of year and so never mate. This is an example of a:

QUESTION 3**Key: D**

A mountain range splits one population into two groups that slowly diverge into separate species. This is:

QUESTION 4**Key: B**

A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

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Synthesis and exit ticket

- Exit claim: explain species concepts using evidence from lab-14_macroevolution.md.
- Must include: Species, Biological species concept, Reproductive isolation.
- Revision prompt: what evidence would change your explanation?

CLAIM

Species concepts

EVIDENCE

lab-14_macroevolution.md

REVISION

What would change your mind?